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Title: "Adaptive Low-Light Image Enhancement Via Self-Paced Learning and Denoising**"**

Abstract

Low-light image enhancement is crucial for improving visual data quality, enabling better performance in subsequent image processing tasks. This project addresses the inherent challenges of restoring images captured under poor lighting conditions. Existing deep learning methods often treat all training samples uniformly, which can impede network convergence, especially during initial training. The SPNet, a novel self-paced learning-based low-light image enhancement network. SPNet incorporates a feature extraction module, a low-light image decomposition module, and a pre-trained denoise module. The network strategically prioritizes the learning of simpler image samples early in the training process, thereby enhancing overall performance. This approach significantly improves image restoration quality. Additionally, SPNet shows quantitative improvements over conventional methods.

Keywords: *Low-light image enhancement (LIE), Self-paced learning (SPL), SPNet, Deep learning, Feature extraction module (FEM).*

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